

FMGE PREP · JUNE 2027

Respiratory System

Complete Study Guide

For: Final-year MBBS · FMGE Aspirants

Coverage: Anatomy · Physiology · Pathology · Medicine · Pharmacology

Estimated Time: 10 days of focused study

If you complete this guide cover to cover, you will own Respiratory in FMGE.

Table of Contents

1. Recommended Resources
2. Anatomy (Day 1)
3. Physiology (Day 2–3)
4. Pathology (Day 4–5)
5. Medicine (Day 6–8)
6. Pharmacology (Day 9)
7. Quick-Fire Revision (Day 10)

Study Order — Do Not Skip Sequence

Anatomy → Physiology → Pathology → Medicine → Pharmacology

Build the foundation first. Mixing subjects randomly will confuse and waste time.

Recommended Resources

Subject	Best Lecture (Marrow)	Reference Book
Physiology	Dr Soumen Manna	Ganong Ch. 34–37
Anatomy	Dr Ashwani Kumar	BD Chaurasia Vol 1
Pathology	Dr Sparsh Gupta	Robbins Ch. 15
Medicine	Marrow + MedCram (free, YouTube)	Harrison Ch. 279–316
Pharmacology	Marrow + Speed Pharmacology (YT)	KD Tripathi Ch. 16

Tutor tip: For FMGE, **Marrow alone is enough**. Use books only when a concept doesn't click after the video.

Part 1 — Anatomy Day 1

Lung Anatomy

- **Right lung:** 3 lobes (upper, middle, lower), 10 segments — wider & shorter.
- **Left lung:** 2 lobes (upper, lower), 8–9 segments — longer & narrower (cardiac notch).
- **Right main bronchus** is more vertical → foreign body lodges here.
- **Carina:** at T4 / angle of Louis — most sensitive cough reflex point.
- **Hilum contents (from posterior → anterior):** bronchus, pulmonary artery (above), veins (below/front).

Pleura

- **Visceral pleura:** no pain fibers (autonomic only).
- **Parietal pleura:** pain fibers — intercostal & phrenic nerves.

- **Costodiaphragmatic recess:** most dependent — fluid collects here first.

Diaphragm — Openings

Opening	Vertebral level	Contents
Vena caval	T8	IVC + right phrenic nerve
Esophageal	T10	Esophagus + vagus (10 letters)
Aortic	T12	Aorta + thoracic duct + azygos

Mnemonic: *I 8 (ate) 10 eggs At 12.*

Muscles of Breathing

- **Inspiration:** Diaphragm (chief), External intercostals; accessory = SCM, scalenes.
- **Expiration:** Passive at rest. Forced expiration = Internal intercostals + abdominals.

Part 2 — Physiology Day 2-3

Lung Volumes & Capacities

$$\text{TLC} = \text{IRV} + \text{TV} + \text{ERV} + \text{RV} \approx 6000 \text{ mL}$$

IRV	= 3000 mL	(Inspiratory Reserve)
TV	= 500 mL	(Tidal Volume – one normal breath)
ERV	= 1100 mL	(Expiratory Reserve)
RV	= 1200 mL	(Residual Volume – cannot be exhaled)

Capacity	Components	Value
IC (Inspiratory)	IRV + TV	3500 mL
FRC (Functional Residual)	ERV + RV	2300 mL
VC (Vital)	IRV + TV + ERV	4600 mL
TLC (Total Lung)	All four	6000 mL

HIGH YIELD: RV, FRC, TLC *cannot* be measured by simple spirometry. Use **helium dilution** or **body plethysmography**.

Obstructive vs Restrictive — The Master Table

	Obstructive (Asthma, COPD)	Restrictive (Fibrosis, Sarcoid)
FEV1	↓↓	↓
FVC	Normal or ↓	↓↓
FEV1/FVC	< 0.7 (low)	> 0.7 (normal/high)
TLC	↑ (air trapping)	↓
RV	↑	↓
DLCO	↓ in emphysema	↓ in fibrosis

Dead Space

- **Anatomical:** conducting airways ≈ 150 mL (measured by *Fowler's* method).
- **Alveolar:** ventilated but not perfused ($V/Q = \infty$).
- **Physiological:** anatomical + alveolar (measured by *Bohr's* method).

V/Q Ratio

- Normal $V/Q = 0.8$.
- **Apex:** $V/Q = 3.3$ — over-ventilated, under-perfused. *TB loves the apex.*
- **Base:** $V/Q = 0.6$ — under-ventilated, over-perfused.
- $V/Q = 0 \rightarrow$ shunt (perfused, not ventilated).
- $V/Q = \infty \rightarrow$ dead space (ventilated, not perfused).

Oxygen Transport

- **98.5% on Hb** · 1.5% dissolved in plasma.
- 1 g Hb carries **1.34 mL O₂** (Hüfner's constant).
- Hb 15 g/dL $\rightarrow$ O₂ capacity $\approx$ **20.1 mL/dL**.

Oxygen Dissociation Curve

Right shift ($\uparrow$ O ₂ release)	Left shift ($\uparrow$ Hb affinity)
$\uparrow$ CO ₂ (Bohr) · $\uparrow$ H ⁺ · $\uparrow$ Temp · $\uparrow$ 2,3-DPG · Exercise	$\downarrow$ CO ₂ · $\downarrow$ H ⁺ · $\downarrow$ Temp · $\downarrow$ 2,3-DPG · HbF · CO poisoning · MetHb

CO₂ Transport

- **70% as bicarbonate (HCO₃⁻)** — dominant.
- 23% as carbamino-Hb · 7% dissolved.
- **Haldane effect:** oxygenation promotes CO₂ release.

Control of Breathing

- **Central chemoreceptors (medulla):** respond to CO₂/H⁺ — main driver.
- **Peripheral (carotid, aortic bodies):** respond to PO₂ < 60 mmHg, $\uparrow$ CO₂, $\downarrow$ pH.
- In **COPD** CO₂-retainers, drive shifts to hypoxia $\rightarrow$ **never give high-flow O₂**.

Surfactant

- Made by **Type II pneumocytes**; main lipid = **DPPC**.
- $\downarrow$ surface tension · prevents alveolar collapse · stabilizes small alveoli.
- Deficiency $\rightarrow$ **IRDS** in preterm babies.
- Laplace: $P = 2T / r$. Without surfactant, small alveoli collapse into large ones.

Part 3 — Pathology Day 4–5

Obstructive Lung Diseases

Emphysema

- Permanent alveolar enlargement + wall destruction (no fibrosis).
- **Centriacinar**: upper lobe, smokers (most common).
- **Panacinar**: lower lobe, **α 1-antitrypsin deficiency** (PiZZ — liver cirrhosis + emphysema).
- **Paraseptal**: subpleural → spontaneous pneumothorax in tall young males.
- Pathogenesis: smoking → neutrophil/macrophage **elastase** ↑ → alveolar destruction.

Chronic Bronchitis

- Clinical: productive cough ≥ 3 mo/yr for ≥ 2 consecutive years.
- **Reid index** > 0.5 (mucous gland hypertrophy).
- "Blue bloater" — cyanosis, edema, hypercapnia, polycythemia.

Asthma

- Reversible airflow obstruction.
- Sputum: **Curschmann spirals** + **Charcot-Leyden crystals**.
- Goblet cell hyperplasia · subepithelial fibrosis · smooth muscle hypertrophy · eosinophils.
- Status asthmaticus → severe attack unresponsive to bronchodilators.

Bronchiectasis

- Permanent abnormal bronchial dilatation.
- Causes: CF (West), **TB (India)**, post-measles/pertussis, **Kartagener** (situs inversus + sinusitis + bronchiectasis).
- Sputum: 3 layers — frothy, mucopurulent, purulent.

Restrictive / Interstitial Lung Disease

Disease	Key Feature
Silicosis	Upper lobe; eggshell calcification ; ↑ TB risk
Asbestosis	Lower lobe; ferruginous bodies ; pleural plaques; mesothelioma
Coal worker's	Progressive massive fibrosis; Caplan (with RA)
Sarcoidosis	Non-caseating granulomas; bilateral hilar nodes; ACE↑; steroids
Hypersensitivity pneumonitis	Farmer's lung, bird fancier's lung — upper lobe
IPF	UIP pattern; honeycombing ; poor prognosis

Lung Tumors

Type	Risk	Location	Notes
Squamous	Smoking	Central	Cavitation · PTHrP → hyperCa · Pancoast
Adenocarcinoma	Non-smoker, women	Peripheral	Most common · EGFR mutations
Small cell (SCLC)	Smoking	Central	SIADH · ACTH · Lambert-Eaton · never surgery
Large cell	Smoking	Peripheral	β-hCG · gynecomastia
Carcinoid	Any	Central	5-HT · octreotide scan

Pleural Effusion — Light's Criteria

Exudate if *any one* is true:

- Pleural protein / serum protein > **0.5**
- Pleural LDH / serum LDH > **0.6**
- Pleural LDH > 2/3 of upper-normal serum LDH

Transudate	Exudate
CCF · cirrhosis · nephrotic · hypothyroidism · Meigs	Pneumonia · TB · malignancy · PE · RA · SLE

Chylothorax: milky; triglycerides > 110 mg/dL; thoracic duct injury.

Part 4 — Medicine

Day 6–8 · Highest FMGE Yield

Asthma Management (GINA)

Step 1: SABA (salbutamol) PRN
Step 2: Low-dose ICS + SABA PRN
Step 3: Low ICS + LABA OR Medium ICS
Step 4: Medium/high ICS + LABA
Step 5: + tiotropium / biologics (omalizumab, mepolizumab)

Acute Severe Asthma

- O_2 to $SpO_2 \geq 94\%$.
- Nebulized SABA + ipratropium.
- IV/oral prednisolone.
- IV $MgSO_4$ (bronchodilator).

DANGER SIGN: Normal or rising $PaCO_2$ in acute asthma = exhaustion / impending respiratory arrest.

Biologics — High Yield

Drug	Target	Indication
Omalizumab	Anti-IgE	Allergic asthma
Mepolizumab / Reslizumab	Anti-IL-5	Eosinophilic
Benralizumab	Anti-IL-5Ra	Eosinophilic
Dupilumab	Anti-IL-4Ra	Severe eosinophilic

COPD

Spirometry: $FEV1/FVC < 0.7$ post-bronchodilator.

GOLD Staging ($FEV1$ % predicted)

- GOLD 1: $\geq 80\%$ (mild) · GOLD 2: 50–79% · GOLD 3: 30–49% · GOLD 4: $< 30\%$.

Stable Treatment (GOLD groups)

A (low sx, low risk): SABA / SAMA PRN
B (high sx, low risk): LABA or LAMA
E (high exacerbation): LABA + LAMA
+ ICS if eosinophils > 300

Acute Exacerbation

- Controlled O₂ (target 88–92%).
- Bronchodilators (SABA + SAMA) + Prednisolone 40 mg × 5 d.
- Antibiotics if purulent sputum (amoxicillin / doxy / azithromycin).
- **NIV (BiPAP)** if pH < 7.35 with ↑ PaCO₂.

LTOT: PaO₂ ≤ 55 mmHg or SpO₂ ≤ 88% → ≥ 15 hrs/day.

Pneumonia

- Most common organism in CAP: **Streptococcus pneumoniae**.
- Atypicals: Mycoplasma, Legionella, Chlamydia — treat with macrolide / doxy / FQ.

CURB-65 (hospitalization)

- C: Confusion · U: Urea > 7 mmol/L · R: RR ≥ 30 · B: SBP < 90 or DBP ≤ 60 · 65: Age ≥ 65
- 0–1 outpatient · 2 hospital · ≥3 consider ICU.

Legionella

- Source: contaminated water (AC, cooling towers).
- **Hyponatremia + ↑ LFT + diarrhea + altered sensorium.**
- Diagnose: **urinary antigen**. Treat: **azithromycin / FQ** (not amoxicillin).

Aspiration pneumonia: right lower lobe (upright) or right upper lobe posterior segment (supine).

Tuberculosis (NTEP — Very High Yield)

Drug	Mechanism	Key Side Effect
Isoniazid (H)	Mycolic acid synthesis	Peripheral neuropathy (give B6) · hepatitis · SLE-like
Rifampicin (R)	RNA polymerase	Orange urine · hepatitis · enzyme inducer
Pyrazinamide (Z)	Active in acidic pH	Hyperuricemia (gout) · hepatitis
Ethambutol (E)	Arabinogalactan synthesis	Optic neuritis (red-green color loss)
Streptomycin (S)	30S ribosome	Ototoxicity (vestibular) · nephrotoxicity

NTEP Regimens

New pulmonary TB: 2HRZE / 4HR
 Retreatment: 2HRZES / 1HRZE / 5HRE
 TB meningitis/bone: 2HRZE / 10HR (12 months total)

DR-TB

- **MDR-TB:** resistant to H + R → Bedaquiline + Linezolid + companions.
- **XDR-TB:** MDR + FQ-R + aminoglycoside-R.

Mantoux (TST)

Population	Positive threshold
Immunocompetent, no risk	≥ 10 mm
HIV / immunosuppressed	≥ 5 mm
IGRA (QuantiFERON)	Not affected by BCG

Pulmonary Embolism

- **Virchow's triad:** stasis + hypercoagulability + endothelial injury.
- Sudden dyspnea + pleuritic chest pain + hemoptysis + DVT signs.
- ECG: **S1Q3T3** (classic); sinus tachy (most common).
- Investigation of choice: **CT pulmonary angiography (CTPA)**.
- **D-dimer:** high sensitivity, low specificity — a *negative* result rules out PE in low-probability patients.
- Echo: RV strain, McConnell sign.

Treatment

- Stable: anticoagulation (LMWH → warfarin / DOAC).
- Massive PE + shock: **thrombolysis (alteplase)**.
- Surgical embolectomy if thrombolysis contraindicated.

Pneumothorax

Type	Cause	Treatment
Primary spontaneous	Tall thin young male; bleb rupture	Small: O ₂ + observe; Large: aspiration / chest drain
Secondary	COPD · asthma · TB · CF	Always chest drain
Tension	One-way valve	Needle decompression at 2nd ICS, MCL — immediately , then chest drain

Do not wait for CXR in tension pneumothorax. Tracheal deviation away, absent breath sounds, hypotension, ↑ JVP → decompress now.

Respiratory Failure

	Type 1	Type 2
PaO ₂	↓ (< 60)	↓
PaCO ₂	Normal / ↓	↑ (> 45)
Mechanism	V/Q mismatch · diffusion	Hypoventilation
Causes	Pneumonia · pulm edema · ARDS	COPD · neuromuscular · obesity
O ₂ therapy	High-flow safe	Controlled low-flow

ARDS — Berlin Definition

- Onset ≤ 1 week of trigger.
- Bilateral infiltrates (not from effusion / collapse).
- **P/F ratio:** Mild 200–300 · Moderate 100–200 · Severe < 100.

Management

- **Lung-protective ventilation:** TV 6 mL/kg IBW.
- PEEP to maintain oxygenation.
- **Prone position** if P/F < 150.
- No routine steroids (except late fibroproliferative phase).

Part 5 — Pharmacology Day 9

Bronchodilators — β 2 Agonists

Drug	Duration	Use
Salbutamol	Short 4–6 h	Acute relief (SABA)
Formoterol	Long 12 h	Fastest-onset LABA
Salmeterol	Long 12 h	Maintenance — slow onset
Indacaterol	Ultra-long 24 h	Once-daily COPD

Anticholinergics (M3 blockers)

- **Ipratropium (SAMA)**: 3–4×/day, more COPD.
- **Tiotropium (LAMA)**: once daily, ↓ COPD exacerbations.

Methylxanthines

- Theophylline — narrow therapeutic index (toxic > 20 µg/mL).
- SE: arrhythmia, seizures, GI upset.
- Erythromycin, cimetidine ↑ levels (enzyme inhibition).

Inhaled Corticosteroids (ICS)

- Beclomethasone · budesonide · fluticasone · ciclesonide.
- SE: **oral candidiasis** (rinse mouth), dysphonia, HPA suppression (high-dose).
- Ciclesonide = prodrug → activated in lung → minimal systemic SE.

Leukotriene Antagonists

- **Montelukast**, zafirlukast — CysLT1 blockers.
- Aspirin-exacerbated asthma · exercise-induced · mild persistent.
- Aspirin-sensitive asthma: arachidonic acid shunted to leukotrienes — avoid NSAIDs.

Mucolytics & Antitussives

- **N-acetylcysteine**: breaks disulfide bonds; also paracetamol antidote.
- Bromhexine · Ambroxol (its active metabolite).
- Guaifenesin — expectorant.
- Dextromethorphan — central NMDA antagonist (antitussive).
- Codeine — opioid antitussive.

Quick-Fire Revision

Day 10

Read this list aloud the night before any test

- Foreign body → **right main bronchus** (more vertical).
- Surfactant → **Type II pneumocyte**; main lipid DPPC.
- RV / FRC / TLC → cannot be measured by spirometry.
- **FEV1/FVC < 0.7** → **obstructive**; ratio normal/high → restrictive.
- Right shift ODC → $\uparrow$ CO₂, $\uparrow$ H⁺, $\uparrow$ T°, $\uparrow$ 2,3-DPG.
- HbF and CO → left shift.
- SCLC → **SIADH, ACTH, Lambert-Eaton**.
- Squamous cell → **PTHrP** → **hypercalcemia**.
- Legionella → urinary antigen + azithromycin.
- Tension pneumothorax → needle 2nd ICS MCL — *before* CXR.
- Isoniazid → peripheral neuropathy → give **pyridoxine (B6)**.
- Rifampicin → enzyme inducer + orange urine.
- Ethambutol → optic neuritis (red-green).
- ARDS → TV 6 mL/kg + prone if P/F < 150.
- V/Q at apex = 3.3 (TB zone); base = 0.6.
- Reid index > 0.5 → chronic bronchitis.
- α 1-AT deficiency → **panacinar** emphysema (lower lobe).
- Light's criteria: protein ratio > 0.5 or LDH ratio > 0.6 → exudate.
- NTEP new TB regimen: **2 HRZE / 4 HR**.
- Massive PE + shock → thrombolysis (alteplase).

Final word from your tutor: One pass of this guide + Marrow videos for any weak topic + 200 MCQs on respiratory = a confident, scoring respiratory section for FMGE. Don't aim for perfection on first read — aim for *recognition*. Revision is where the marks come from.